

Unit 6, Lesson 4: Exploring Slope

Benchmark

MA.8.AR.3.2 Given a table, graph, or written description of a linear relationship, determine the slope.

Learning Targets

- ☐ I can define slope.
- ☐ I can determine the slope given two points.

6.4.1 Warm-Up: Convince the Class

- 1. Write an explanation to convince the class whether this is a math fail or a reasonable approximation.

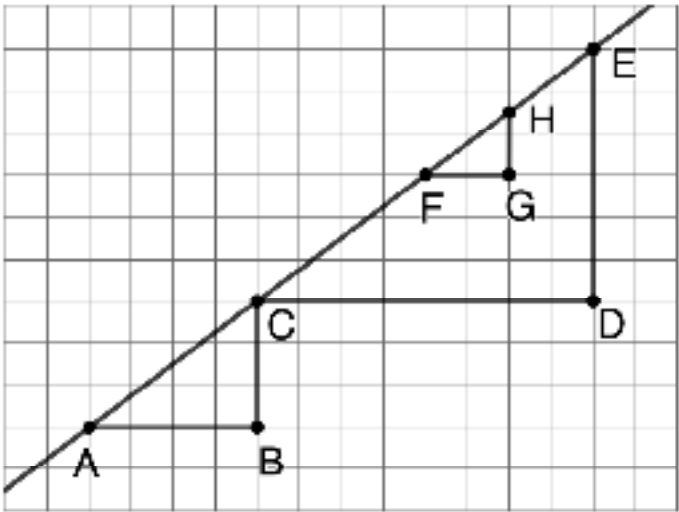


6.4.2 Exploration: Exploring Slope Through Ratios

The figure shows three right triangles, each with its **hypotenuse** on the same line. Each gridline is equal to one unit.



The **hypotenuse** is the longest side of a right triangle; the side opposite the right angle.



- 1. Complete the table.

Triangle	Vertical Side Length	Horizontal Side Length	$\frac{\text{vertical}}{\text{horizontal}}$	Simplified Ratio
<i>ABC</i>			$\frac{\square}{\square}$	
<i>CDE</i>			$\frac{\square}{\square}$	
<i>FGH</i>			$\frac{\square}{\square}$	

2. What do you notice about the simplified ratios for the three triangles?
3. Discuss with your partner and write a conjecture about the ratio of the vertical side length to the horizontal side length of any right triangle with a hypotenuse on the same line as the triangles above.

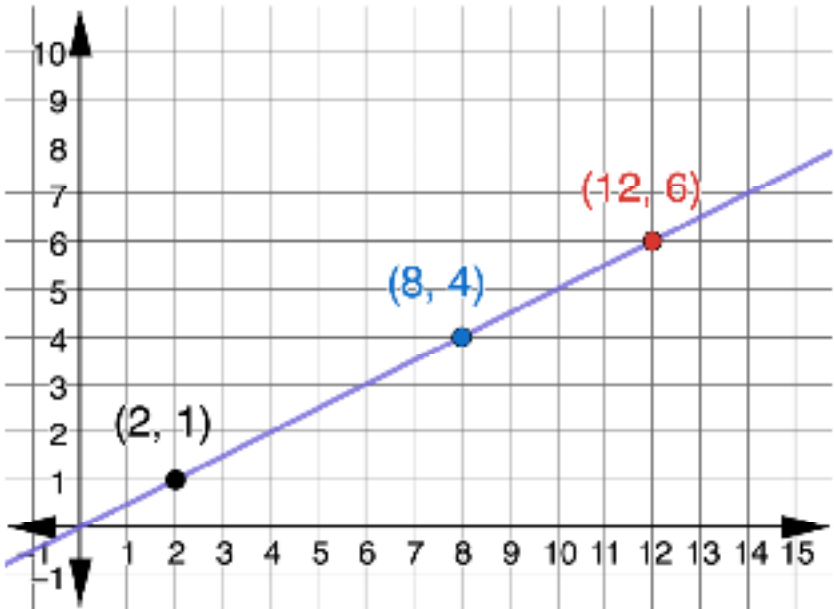
The ratio of the vertical side length to the horizontal side length of a right triangle is known as the **slope** of the hypotenuse.



The **slope** is the ratio of the change in the vertical direction (y direction) to change in the horizontal direction (x direction), often expressed as $\frac{\Delta y}{\Delta x}$.

6.4.3 Collaboration: Constant of Proportionality and Slope

1. The graph of a relationship is shown.



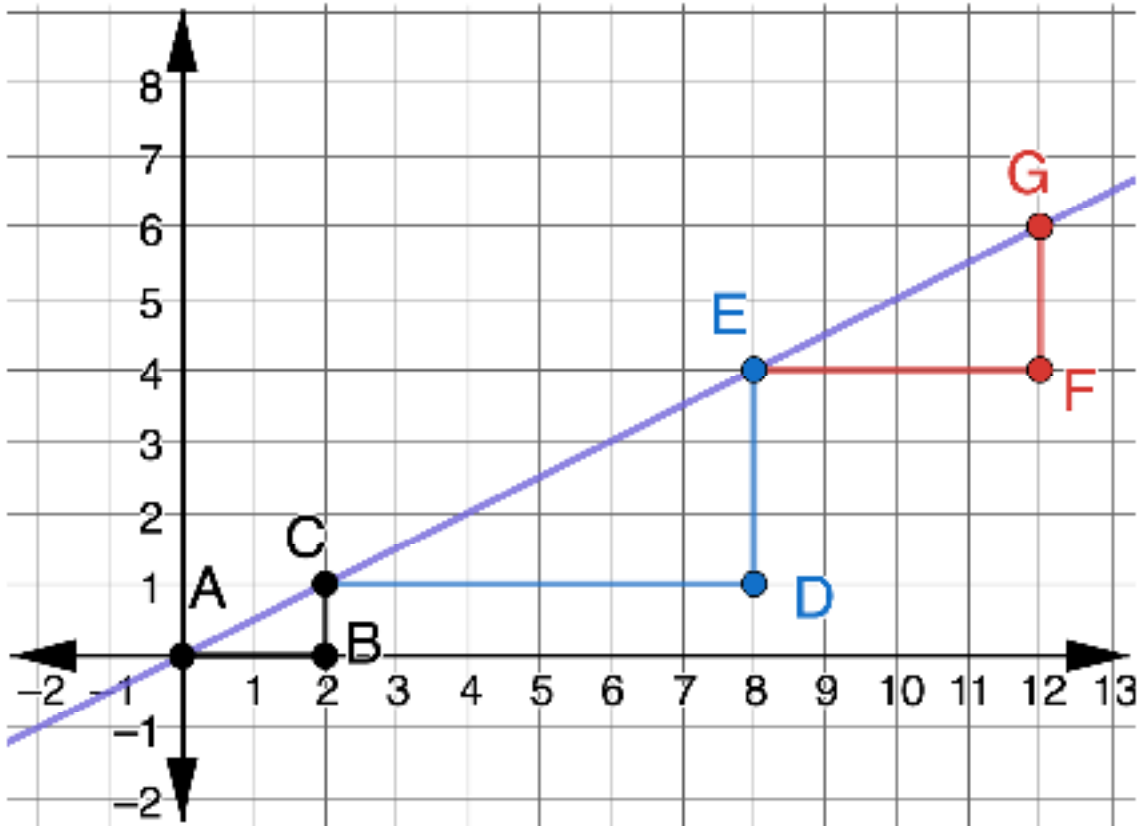
- a. Use the graph to complete the table.

x	y	Ratio $\frac{y}{x}$
2	1	
4		
8		
10		
12		
16		

- b. Use the table to complete the equation with the constant of proportionality.

$y = \boxed{} x$

2. The graph of the same relationship is shown with three right triangles such that their hypotenuses are on the line.

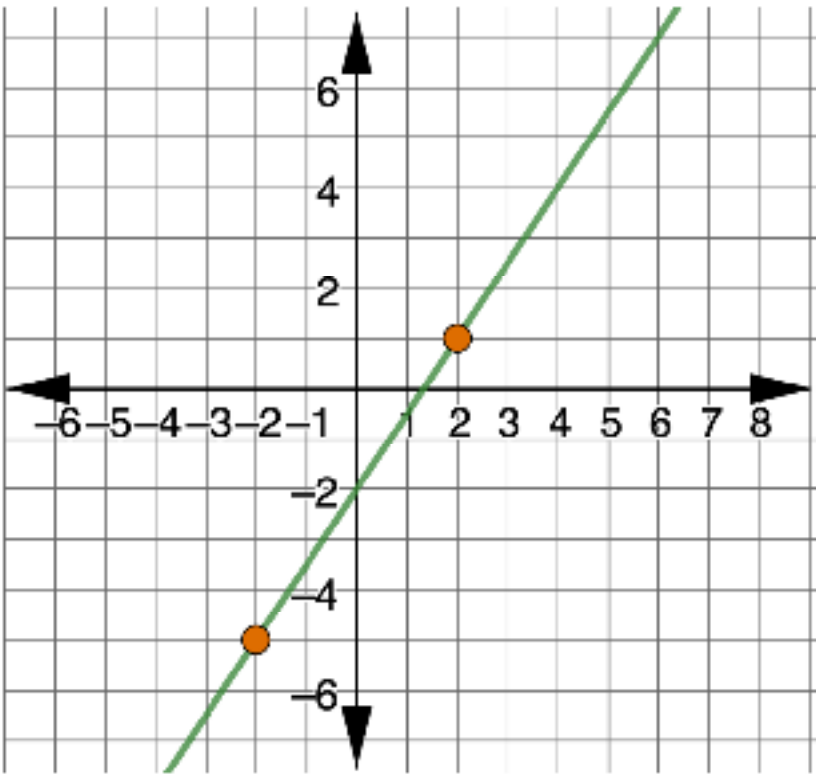


- a. Complete the table for the ratio of the vertical side lengths to the horizontal side lengths of each right triangle.

Triangle	Vertical Side Length	Horizontal Side Length	$\frac{\text{vertical}}{\text{horizontal}}$
ABC	1	2	
CDE		6	
EFG	2		

- b. What is the value of each ratio in simplest form?
- c. What is the slope of the line?
3. Discuss with your partner how identifying the constant of proportionality is related to determining the slope of a proportional relationship. Summarize your discussion.

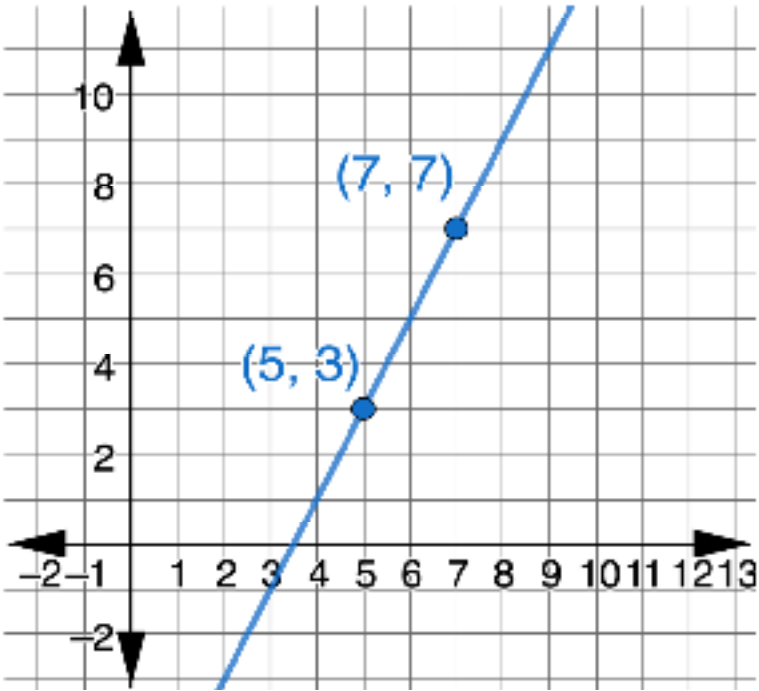
The slope of any linear relationship can be determined using two points on the line.



4. Discuss with your partner how the slope of any line could be determined when right triangles are not given on a graph. Then, summarize your discussion.

6.4.4 Guided Instruction: Finding Slope Given Two Points

1. The graph of a line is shown with the coordinates of two of its points labeled.



- a. Sketch a right triangle with a hypotenuse on the line and vertices connecting the two given points.
- b. Compare your sketch to your partner’s sketch.
- c. Use the right triangle to complete the table.

Vertical Side Length	Horizontal Side Length	$\frac{\text{vertical}}{\text{horizontal}}$	Slope

2. Complete the statement.

- ☐ Only one triangle

☐ More than one triangle

is needed to determine the slope of the line

☐ always the same

☐ sometimes different

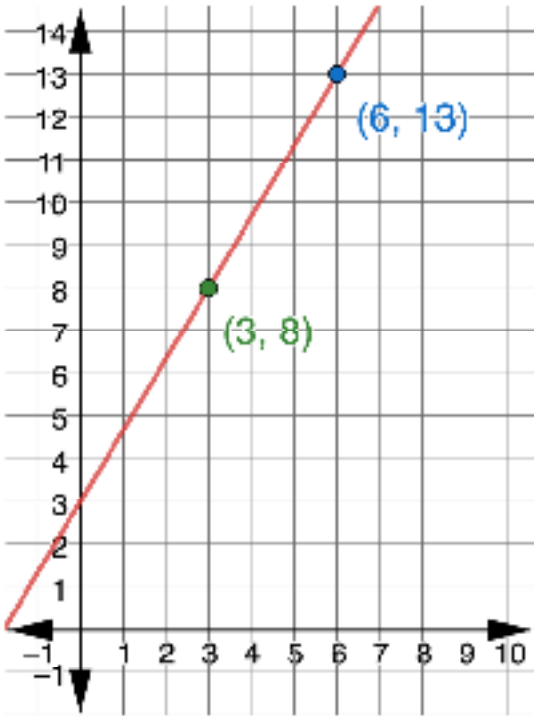
because the slope is

between any two points on the

same line.

3. The graph of a line is shown.

a. Use the graph to complete the table.

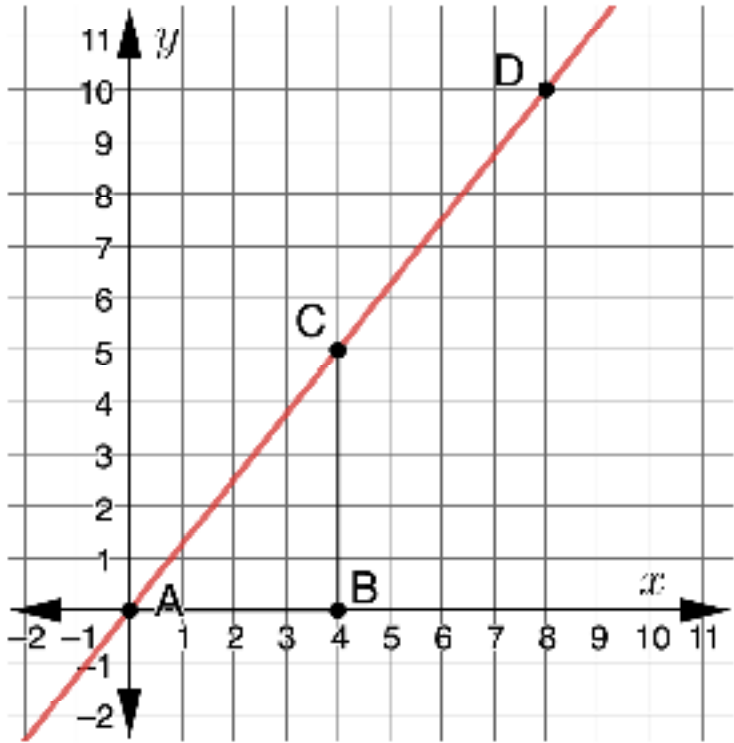


Vertical Side Length	Horizontal Side Length	$\frac{\text{vertical}}{\text{horizontal}}$	Slope

- b. Without counting the units on the graph, how can the vertical distance between two points be determined?
- c. Without counting the units on the graph, how can the horizontal distance between two points be determined?

6.4.5 Your Turn!

1. A line is shown in the coordinate plane.

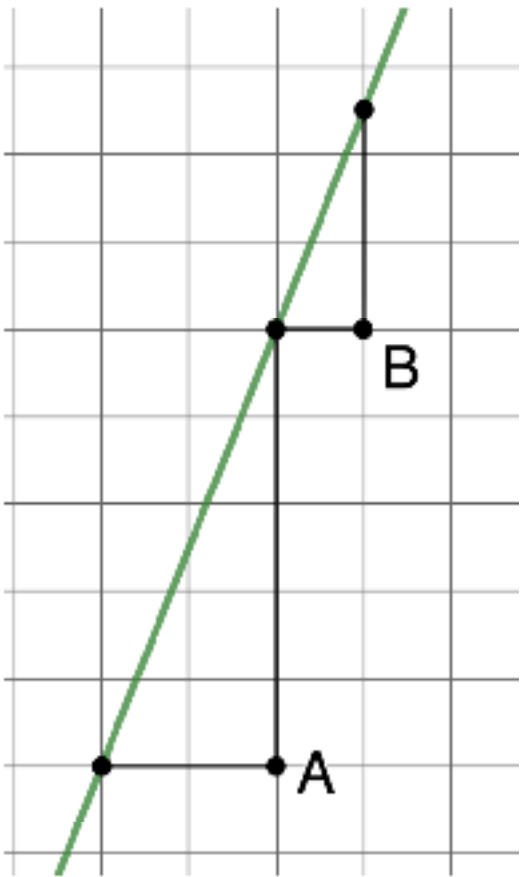


- a. What are the coordinates of C and D ?

Point	Coordinates
C	
D	

- b. Determine the slope of the red line.

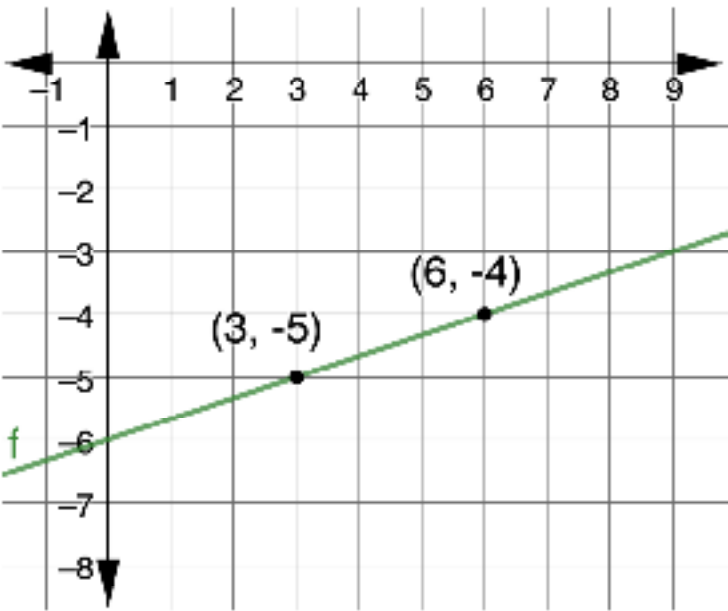
2. Two triangles are shown on the graph such that their hypotenuses lie on the same green line.
- Kiana uses Triangle *A* and says the slope of this line is $\frac{5}{2}$. Jeff uses Triangle *B* and argues that the slope of the green line is 2.5.
- Explain who is correct.



6.4.6 Wrap-Up: Error Analysis

1. The graph of a line is shown with the coordinates of two points labeled.

Judah found the slope of the line to be 3.



- a. Explain the error Judah made.
- b. Determine the correct slope of the line.

